



Title of Report A possible solution for milk tea shops to find a privileged location

Score _____

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Content

1. The overall view of the research.....	3
1.1 The background of the research.....	3
1.2 Market analysis of hot pot and milk tea.....	4
1.2.1 The size of the market.....	4
1.2.2 The development trend.....	5
1.2.3 Competitors Analysis.....	6
1.3 Purpose of research.....	6
1.4 Meaning of research.....	7
1.5 The Thinking Framework.....	7
2. The design of survey.....	7
2.1 Survey plan design.....	8
2.2 Model testing.....	8
2.3 Questionnaire and scale design.....	9
2.4 Data processing.....	12
2.4.1 Coding correction.....	12
2.4.2 Variable assignment.....	12
2.4.3 Validity test.....	12
3. The current condition of 'hot pot& milk tea mode'.....	13
3.1 Behavior of consumers.....	13
3.2 "Hot pot + Milk tea" Market SWOT Analysis.....	15
3.3 Hotpot market customer group consumption motivation and market demand analysis.....	15
3.3.1 Customer consumption motivation exploration based on factor analysis.....	15
3.3.2 Analysis of customer consumption willingness based on multiple and orderly Logistic regression.....	17
3.3.3 Variable simulation.....	17
3.3.4 Model results.....	18
4. Models for research.....	19
4.1 API map crawler.....	19
4.2 frequency distribution.....	22
4.3 Linear Regression model.....	23
4.4 Marginal cost & marginal model for saturation.....	24
5. Conclusion.....	25
References.....	27
Appendix.....	28

1. The overall view of the research

1.1 The background of the research

(1) The rapid development of the hot pot industry and the milk tea industry

In recent years, the hotpot industry has been booming. Data show that the market size of China's hotpot industry is nearly 500 billion yuan in 2021, and is expected to reach 668.9 billion yuan in 2025. In addition to 2020, the market size of the hotpot industry has maintained a growing trend. Hot pot, as the most popular dish, has easy standardization, high freedom, fantastic tastes, and social functions, solving the long-time pain point of the catering industry. At the same time, hot pot has thousands of years of culture, has been recognized as suitable for the party chat diet, and hot pot bottom in various forms, not subject to the age level of constraints, in the public consumption blew a red whirlwind. A variety of web celebrity hot pot restaurants emerge in an endless stream, Haidilao hot pot, Xiabu Xiabu, Ba Nu hotpot, Xiaolongkan hot pot and other hot pot brands are also rapidly emerging.

At the same time, milk tea shops are also very popular in recent years, and chain brands such as Mixue Ice Cream & Tea, Coco, Yi Diandian, Cha Baidao and so on are very popular across the country. Under the strong market demand, existing milk tea shops, especially high-end existing milk tea shops, have emerged in China, and the tea market has achieved rapid growth.

Through franchise mode, more franchisees participate in milk tea industry, and expand the whole market. By the end of 2020, there were about 348,000 existing milk tea shops in China. According to the retail consumption value, the market size of domestic existing tea shops was only 800 million yuan in 2015, and by 2020, Chinese people had drunk 66.2 billion yuan annually. From 2022 to 2027, the in-depth research report of 2027 is expected to reach 211 billion yuan by 2025.

Milk Tea Industry Market Size and forecast in China, 2013-2024

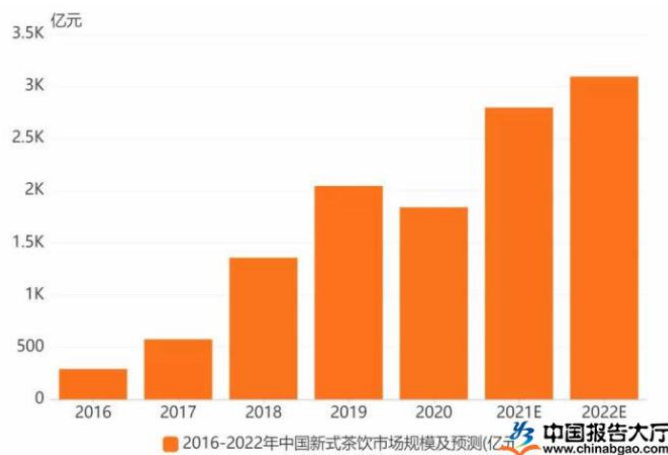


Chart 1 source: IMedia Data Center

(2) Hot pot consumers' drink demand

Many consumers always like to eat while drinking drinks, this phenomenon is more common in hot pot restaurants. Many consumers will have the demand for drinks when eating hot pot. It is not difficult to find that almost all the hotpot restaurants have the supply and sales of drinks, and even some hot pot brands have even created their own beverage stores. However, many customers choose to buy drinks in milk tea shops, so as to bring their own hot pot restaurants, such a phenomenon is very common. We can even notice that some hotpot restaurants and milk tea shops are very close together.

1.2 Market analysis of hot pot and milk tea

1.2.1 The size of the market

The market size of the hot pot and milk tea industry will continue to grow in the future.

The market size of the hot pot industry in China reached 499.8 billion yuan in 2021 and is expected to reach 668.9 billion yuan in 2025, according to the data. Apart from 2020, the market size of the hot pot industry has maintained an increasing trend. The hot pot industry is easier to scale and standardize than other catering industries.

Hot pot Industry Market Size and forecast in China, 2013-2024

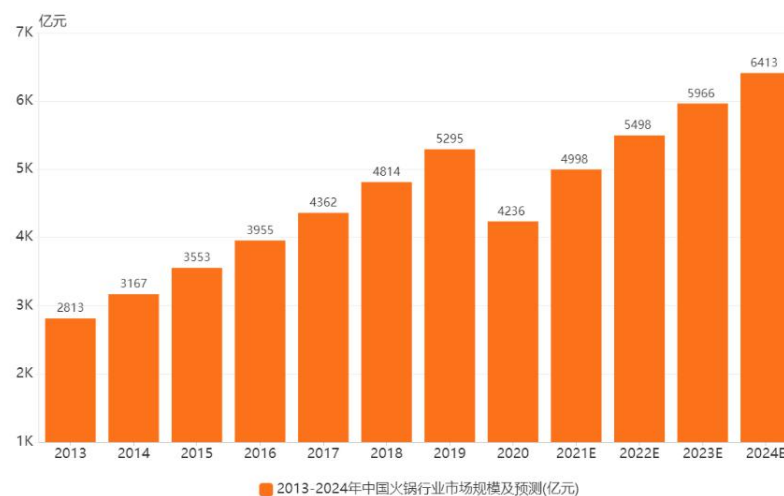


Chart 2 source: IMedia Data Center

The market size of milk tea in China in 2021 was 279.59 billion yuan, according to the data. Iterations in Internet technology and improvements in online payment technology have made online shopping more convenient, effectively easing the loss of offline operations of the milk tea industry during the pandemic. The acceptance of the milk tea consumer group has gradually increased, and the category has continuously expanded and innovated. Consumers' enthusiasm for milk tea continues to rise, and demand has risen accordingly. The market size of milk tea is estimated to reach 374.93 billion yuan in 2025. Nowadays, the milk tea shops are adopting franchise mode in China. Shoppers who want to take part in can get the franchise and take full

responsibilities. This mode has helped milk tea shops expand rapidly in recent years. By the end of 2020, there were about 348,000 existing milk tea shops in China.

Milk Tea Market Size and Forecast in China, 2016-2025 (100 million Yuan)



Chart 2 source: IMedia Data Center

1.2.2 The development trend

The heterogeneous hot pot industry will be a long-term trend.

There is a higher degree of homogeneity among the various hotpot restaurants, which have not been able to compete for a long time..Hot pot catering businesses should find their own positioning, occupy a segment category or seek innovation in the operation, so as to be able to successfully "out of the circle". Seeking heterogeneity will be a long-term trend in the development of the hot pot catering industry.

The focus on health and safety and food quality management mode is a general trend. What has changed is brand positioning and management patterns, and what has not changed is the pursuit of health and safety, service attitudes and food quality..With the improvement of people's living quality and income level, people will pursue "quality" rather than "quantity" in catering consumption. In the future, the hotpot restaurant business will form a complete set of health, food quality and safety management systems and service tracking systems.

Catering to the needs of consumers is the only way for the hot pot industry to develop and innovate. In the past, most hot-pot restaurants stayed in business by improving store decor, inviting celebrities to ring the bell and other ways to generate heat. These methods bring temporary heat to hotpot restaurants, but the external objects fail to impress the consumer. Hot pot to "milk tea + hot pot" business model to win the favor of youthful consumers, hot pot takeout, self-heating tiny hot pot to meet the youthful consumers in different situations as much as possible to eat satisfactory hot pot needs. Times are changing and so is the pursuit of consumers in hotpot consumption, and hotpot enterprises should take the initiative to cater to current hotpot

consumption concepts and make innovations to achieve better development.

1.2.3 Competitors Analysis

(1) Hot pot brand of self-run drinks

Almost all hot pot restaurants will have drinks on the menu, and the menu is also more and more abundant drinks, including drinks, drinks, ice cream and other types. Even many web celebrity stores are famous for their ice cream and have attracted many consumers.

At the same time, there are many hotpot restaurants to carry out milk tea activities or self-derived milk tea shops. For example, in 2019, Haidilao launched four self-research and test milk tea models, including milk tea and jasmine milk green. In 2021, Haidilao not only launched a DIY homemade milk tea shop of 9.9 yuan, but also opened a Haidilao brand milk tea shop in Shanghai and Nanjing. Another example, Xiaolongkan launched "dragon small tea", is officially into the tea market. Earlier before the together, has been the "hot pot + tea" mode.

These self-run drinks of hot pot brands have great advantages in the convenience degree such as distance and time, and even add innovative elements to attract many consumers. In the "hot pot + milk tea" market, these hot pot brands of self-run drinks market competitiveness is very strong.

(2) Other brand milk tea shops

Since 2007, the Chinese mainland milk tea market has entered a stage of rapid development. With the hints of people's economic level and quality of life, the consumption power is also constantly rising, and it plays a vital role in the development of the milk tea industry. Then the market emerged above many milk tea brands, such as, Coco, Lele tea, Cha Baidao and so on. The originally restless milk tea market soon attracted the birth of a large number of new brands, setting off a new war in China's milk tea industry. However, new entrants in the milk tea industry brings new production capacity, new resources at the same time, will hope to be existing enterprises in the market, it may happen with the existing milk tea brand raw materials and market share competition, eventually lead to the existing enterprises in the industry profitability, serious and may endanger the survival of these enterprises.

1.3 Purpose of research

In view of the above phenomenon, we will further conduct a market research around the contact between milk tea shop and hot pot shop to study the site selection of milk tea shop. Our research mainly includes the following parts.

(1) For the group of consumers in the hot pot market, study the demand of consumers buying milk tea while eating hot pot. Since milk tea shops adopt franchise mode, how to locate for a new shop might be a key problem and a practical problem. Our team is going to discover how the distance between the milk tea shop and the hot pot shop will affect the operating profit of the milk tea shop, how to locate the milk tea shop according to the location of the

hot pot shop, how to layout the number and location of the milk tea shop, in order to obtain the maximum profit.

(2) Analyze the current market, understand the current situation of the market and the future development potential, explore the hot pot market and milk tea market consumption status and consumption motivation, find the correlation between the two, explore all kinds of consumption motivation on the influence of consumers consumption hot pot and milk tea intention mode, and based on their consumption motivation to do further analysis.

(3) Classify and analyze the customer groups of the hot pot and milk tea market, understand the characteristics, consumption habits and preferences of this group, and change and innovate the marketing strategies for this group.

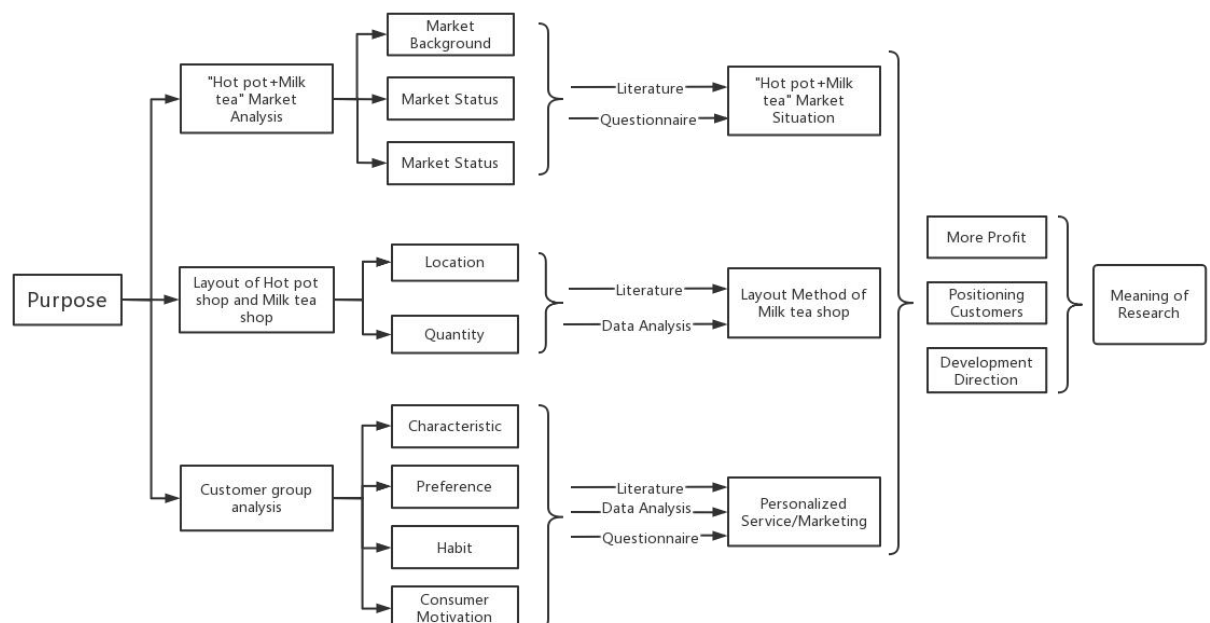
1.4 Meaning of research

(1) Based on the contact between hot pot and milk tea market, help milk tea shops to choose the best location and method of layout.

(2) Positioning customers, finding more consumer groups for milk tea shops, and improving the innovation of operation mode and publicity according to the characteristics of this group, in order to increase the operating profit of milk tea shops.

(3) Help milk tea shop to find the new direction of development in the future and new business opportunities.

1.5 The Thinking Framework



Picture 1

2. The design of survey

2.1 Survey plan design

(1) the subject of investigation

Overall target: People aged 10-70 in all regions of the country who are willing to eat hot pot and drink milk tea in their daily life.

(2) The contents of the investigation

Firstly, understand the hot pot and milk tea consumption market status quo and consumption motivation analysis.

Through descriptive statistics of questionnaire data, this paper analyzes consumers' cognition of hot pot and milk tea consumption market, willingness to consume hot pot and milk tea, consumption motivation and willingness to pay and other information. Through the analysis of hotpot and milk tea consumers' potential consumption motivation, product value perception and other multiple factors, to explore the influence mechanism of each potential motivation on consumers' hotpot and milk tea consumption intention, and based on the motivation analysis of consumers' hotpot and milk tea consumption intention to make further prediction.

Secondly, understand the characteristics of potential customer groups and conduct group portrait analysis. Based on the comprehensive analysis of respondents' hot pot and milk tea consumption intention and subjective perception factors, the group clustering of potential customers for hot pot and milk tea consumption was realized, and the consumption potential of each customer group was measured. Further, by analyzing the group portraits of the two types of customers with higher potential value, a clearer customer value positioning can be obtained. It also facilitates the positioning and differentiation analysis of customer characteristics in the follow-up research.

Third, understand the usage habits and demand preferences of potential customers to achieve precision marketing.

(3) Method of investigation

Our team design questionnaire online and post it to the WeChat Moment. Using the sample to estimate population, we may have an overall view of people's preference of drinking milk tea while eating hot pot, in all folks of life across the country. By using the method of map crawler and collect, we study the POI of milk tea shops and hot pot shops, and build models to see their correlations.

2.2 Model testing

We used the goodness-of-fit to test whether the expected proportion varied significantly from the actual proportion, and whether the collected data fit into the actual situation. Significance above 0.5, indicating that the difference between the sample and the population is due to the influence of systematic rather than accidental factors, so that we can make the inference of our selected factors without external influence. The goodness-of-fit significance obtained from this group of data was 0.867, over 0.5, indicating that the significance was strong and the influencing factors were not accidental.

拟合优度			
	卡方	自由度	显著性
皮尔逊	90.914	107	.867
偏差	90.205	107	.879

Table 1

2.3 Questionnaire and scale design

According to the purpose and content of the survey, the questionnaire of this survey is mainly divided into three parts. That are, the basic information about consumer characteristics, behavior of customers, and customers' motivation to purchase cups of milk tea.

2.3.1 Basic information about consumer characteristics

77.61% of respondents were female and 22.39% were male.

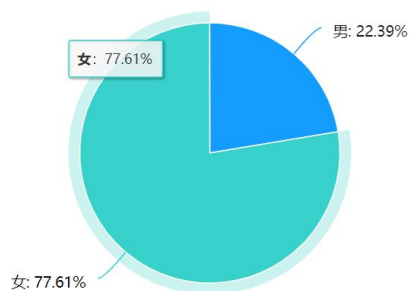


Chart 3

The respondents are from various places. Most of respondents from Shanghai, following with Beijing, while the number of respondents from southern, and western part of China. Tianjin accounts for the least of proportion. It shows the variety of locations.

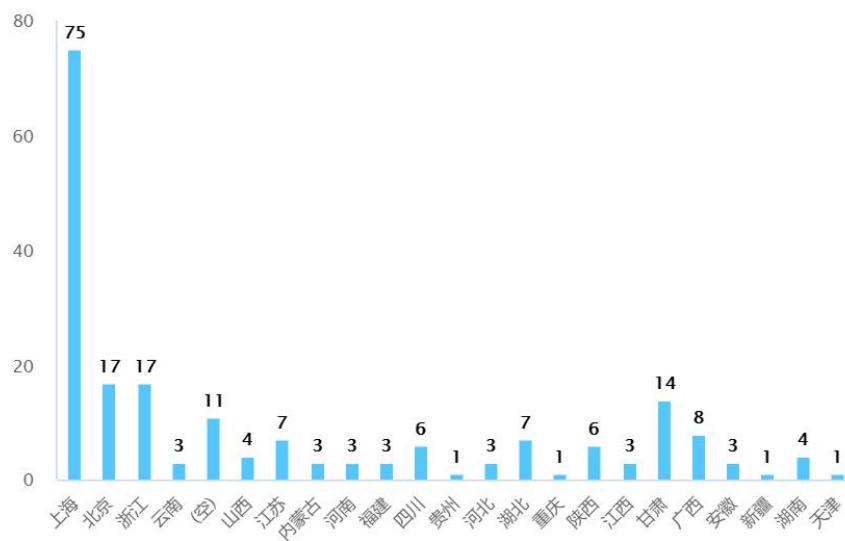


Chart 4

This interview include people from different aging period, in order to see the demand of different ages and target market.

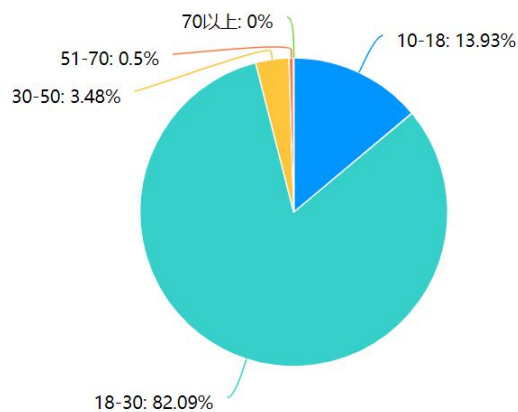


Chart 5

Interviewees are more willing to eat spicy hot pot from Sichuan province, which accounts for 77.61% of respondents in this questionnaire, while Suzhou-Hangzhou chrysanthemum hot pot accounts for the least, only 4.98%.

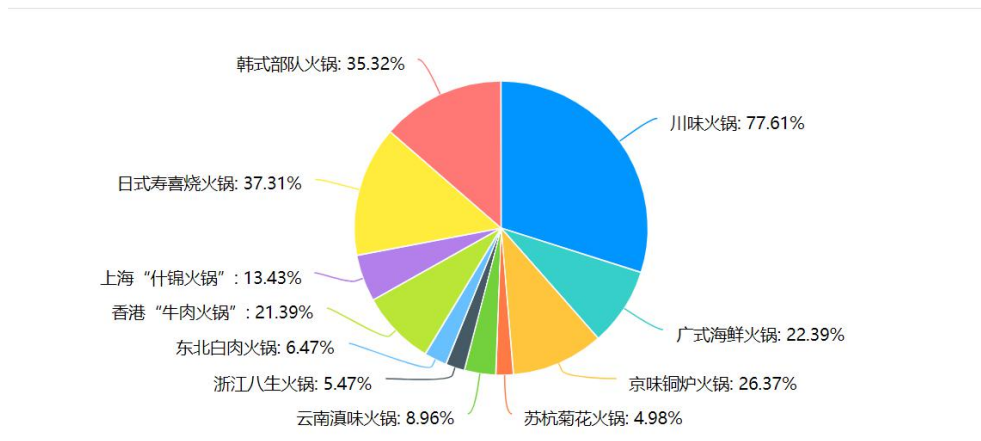


Chart 6

There are 57.71% of respondents in this survey sometimes have the habit of eating hot pot, while no one never eats hot pot. Also, the relative frequency that interviewees eat hot pot once a month is nearly 40%, the frequency that interviewees eat hot pot 3 times a month is nearly 25%. It seems that hot pot is popular among ages.

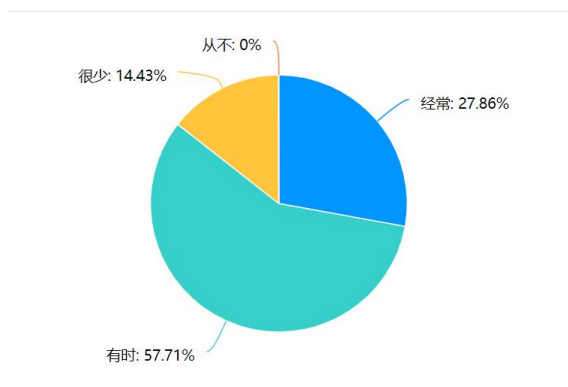


Chart 7

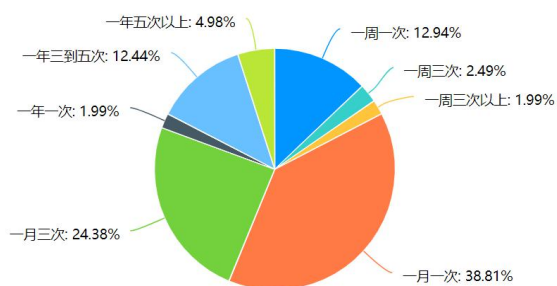


Chart 8

In terms of basic information, our team found that people of different sex, age and region of hot pot and tea consumption may have a different attitude, so choose the above indicators at the same time, considering in terms of

consumer behavior, mainly from the respondents understanding of hot pot and milk tea consumption and participation, the cause of the hotpot, and milk tea consumption, some milk tea intend to explore the three aspects.

2.4 Data processing

2.4.1 Coding correction

In the variable assignment, there will be situations where the assignment does not match the degree, such as the frequency selection in this survey, the computer automatically assigns 1~4 to frequent ~ rare, at this time we need to code and correct the variable assignment to match each value with the content.

2.4.2 Variable assignment

Argument meaning	assignment
gender	Male=1, Female=2
age	10~18=1, 18~30=2, 30~50=3, 51~70=4
Hot pot flavor (multiple choice)	No=0, yes=1
Eat hot pot frequency	Rarely = 1, sometimes = 2, often = 3
Whether it is accompanied by the habit of drinking milk tea	Never=1, rarely =2, sometimes =3, often =4
Reasons for drinking milk tea (multiple choice)	No =0, =1 selected
Pickup habits	Self-pick-up = 1, delivery = 2
Distance (multiple selection)	No =0, =1 selected
Whether to increase the desire to drink milk tea	Yes=1, No=2

Table 2

2.4.3 Validity test

In order to justify the design of the questionnaire and its applicability to this study, the validity of the questions set in the questionnaire that meet the conditions of the scale are tested and analyzed. For structural validity, KMO and Bartlett spherical tests were used to analyze whether they were suitable for factor analysis. According to the test results, its KMO value is 0.633, which is greater than the test coefficient standard of 0.6; and the significance of the Bartlett spherical test is 0.000, and the Bartlett spherical test indicates that the data can be analyzed by factors.

KMO 和巴特利特检验		
KMO 取样适切性量数。		.633
巴特利特球形度检验	近似卡方	314.080
	自由度	91
	显著性	.000

Table 3

3. The current condition of 'hot pot& milk tea mode'

3.1 Behavior of consumers

Our team use cross analysis to the influence of ages to the habit of drinking milk tea while eating hot pot in the seventh question. People aged 10-18 accounts the largest proportion of always drinking milk teas while eating hot pot, the percentage is 32.14%. People aged 18-30 have largest proportion of drinking milk tea sometimes. Since many fruit or ingredients are favored by young people, and teenagers prefer sweet things, milk tea is more popular among young people. On the other hand, none of person above 30 years old always or sometimes drink milk tea while eating hot pot. Nearly half of interviewees in this age period never drink milk tea while eating hot pot. They may consider more about health and balanced diet, and the may not get familiar with the taste of milk tea. Hence, people aged 10-30 have a greater demand for milk tea when eating hot pot, and can be seen as a target market.

X\Y	经常	有时	很少	从不	小计
10-18	9(32.14%)	9(32.14%)	7(25%)	3(10.71%)	28
18-30	43(26.06%)	57(34.55%)	30(18.18%)	35(21.21%)	165
30-50	0(0.00%)	0(0.00%)	4(5.714%)	3(4.286%)	7
51-70	0(0.00%)	0(0.00%)	0(0.00%)	1(100%)	1
70以上	0(0.00%)	0(0.00%)	0(0.00%)	0(0.00%)	0

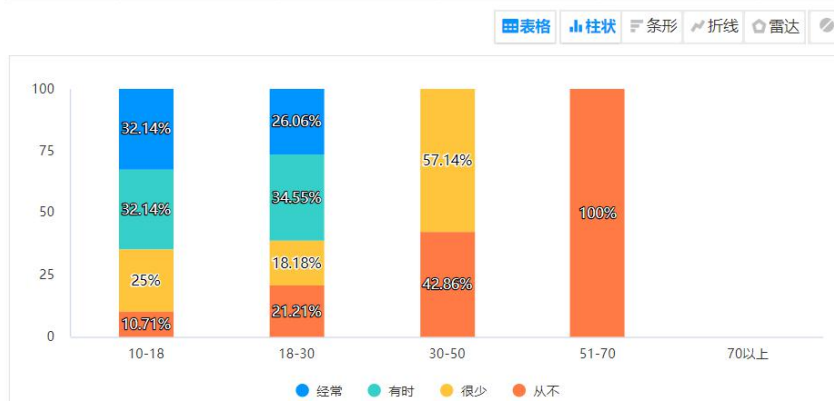


Chart 9

From people who purchase cup of milk tea while eating hot pot, the reasons for drinking milk tea account for the most, that is, quenching thirst and relieving heat and

good taste, accounting for 62.26% and 62.89%, respectively. Another reason, high cost performance, accounts for 10.69% at the lowest. Therefore, we may infer that people have huge demand to release piquancy when they are eating hot pot, especially spicy hot pot.

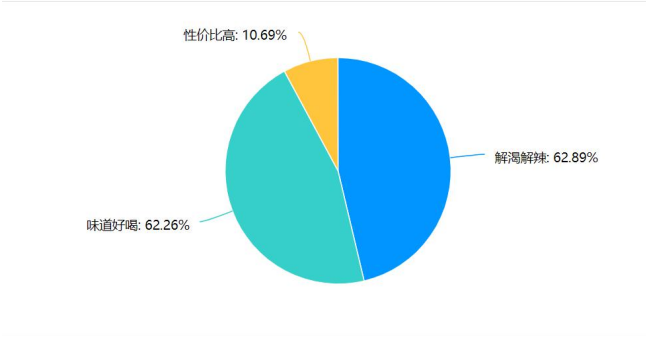


Chart 10

According to the tenth question, the preference is to go to the store or to fetch by oneself, and the eleventh question, when the milk tea shop is located in the range of multiple distances from the hot pot shop, the cross-analysis of the two questions shows that the desire of ordering milk tea to fetch by oneself is greater within 500m and within the range of walking for five minutes. As shown in the figure, according to the analysis of sample recovery data, For one sample, the store next door and within 10km is selected at the same time, and the previous problem chose not to order milk tea when eating hot pot. This is a contradictory and conflicting option, so it can be determined that the sample is a singular value and needs to be removed, and the sample data will not be considered for calculation.

X\Y	到店自取	外卖	小计
隔壁店铺	60(57.14%)	45(42.86%)	105
楼下或者楼上	74(60.16%)	49(39.84%)	123
步行5分钟以内	59(56.19%)	46(43.81%)	105
1km以内	18(69.23%)	8(30.77%)	26
2km以内	3(37.5%)	5(62.5%)	8
5km以内	0(0.00%)	0(0.00%)	0
10km以内	1(100%)	0(0.00%)	1

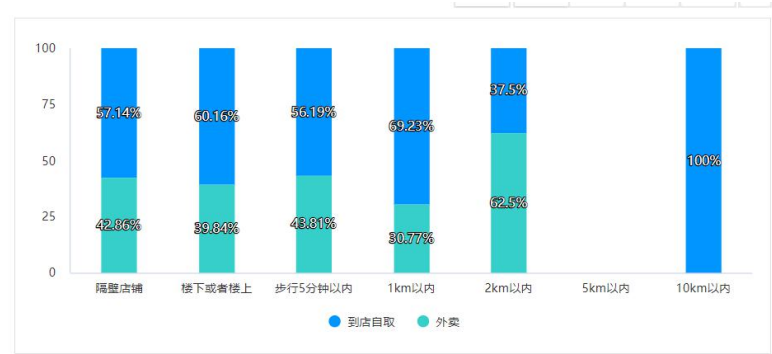


Chart 11

Lastly, we see that nearly 80% of interviewees will choose to purchase a cup of milk tea that they would have not purchased, if there is a milk tea shop near a hot pot shops, no matter the taste of hot pot they prefer, and the price of milk tea they prefer.

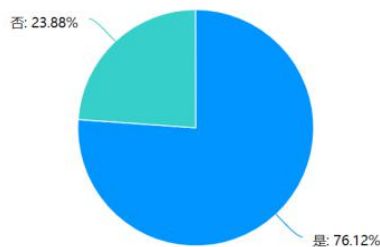


Chart 12

3.2 "Hot pot + Milk tea" Market SWOT Analysis

S (Strengths) 1. Cost-effective, good quality and cheap 2. Rich variety, good taste 3. Keep up with the fashion trend, consumers' desire to buy is high	W (Weaknesses) 1. Customers need to take their own milk tea 2. Seasonal consumption of milk tea is much obvious
O (Opportunities) 1. Consumer demand is large, and the market is vast 2. Hot pot market and milk tea market are hot, with many consumers and a large number of potential customers and target groups	T (Threats) 1. In terms of distance and convenience than drinks in hot pot restaurants 2. There are many competitors, and a large number of milk tea shop brands are rising rapidly, 3. Milk tea production threshold is low, many hot pot restaurants can make their own similar drinks

Table 4

3.3 Hotpot market customer group consumption motivation and market demand analysis

3.3.1 Customer consumption motivation exploration based on factor analysis

Factor selection

The analysis results are shown in the table, in which the first column contains 18 principal components, the corresponding "eigenvalues" in the second column represent the magnitude of the explained variance, and the third and

fourth columns respectively represent the ratio of the variance contained in this component to the total variance, and the cumulative 100%. In general, components with "eigenvalues" greater than one are chosen as principal components.

As can be seen from the table below, the eigenvalues of components 1, 2, 3, 4, 5, 6 and 7, corresponding the taste of spicy hot pot from Sichuan province, sukiyaki from Japan, Mixed meat hot-pot from Shanghai, sometimes drinking milk tea while eating hotpot, releasing spicy and thirst, good taste, cost-effective, are all larger than one and can be judged as the dominant factors. The first component accounts for 14.425% of the total variance and 14.425% of the cumulative variance; The second component accounts for 12.024% of the total variance and 26.449% of the cumulative variance explained. The third component accounts for 9.354% of the total variance and 35.803% of the cumulative variance explained. The fourth component accounted for 7.701% of the total variance, and 43.504% of the cumulative variance. The fifth component accounted for 6.621% of the total variance, 50.125% of the cumulative variance; The sixth component accounts for a total variance of 6.099% and a cumulative variance of 56.233%. The seventh component accounted for 5.780% of the total variance and 62.003% of the cumulative variance. It can be seen that the total explanatory power of the seven factors is excellent. Thus, seven main factors can be obtained by factor analysis to reduce the dimensionality of the scale variables.

总方差解释									
成分	初始特征值			提取载荷平方和			旋转载荷平方和		
	总计	方差百分比	累积 %	总计	方差百分比	累积 %	总计	方差百分比	累积 %
1	2.597	14.425	14.425	2.597	14.425	14.425	2.079	11.552	11.552
2	2.164	12.024	26.449	2.164	12.024	26.449	1.956	10.865	22.418
3	1.684	9.354	35.803	1.684	9.354	35.803	1.809	10.048	32.466
4	1.386	7.701	43.504	1.386	7.701	43.504	1.362	7.569	40.035
5	1.192	6.621	50.125	1.192	6.621	50.125	1.326	7.366	47.401
6	1.098	6.099	56.223	1.098	6.099	56.223	1.314	7.302	54.703
7	1.040	5.780	62.003	1.040	5.780	62.003	1.314	7.300	62.003
8	.941	5.230	67.233						
9	.836	4.646	71.880						
10	.829	4.606	76.486						
11	.708	3.932	80.418						
12	.648	3.598	84.016						
13	.634	3.524	87.541						
14	.581	3.228	90.768						
15	.561	3.115	93.883						
16	.452	2.514	96.397						
17	.388	2.155	98.552						
18	.261	1.448	100.000						

提取方法：主成分分析法。

Table 5

The results of the lithotripsy map are shown in Figure. The number of components on the horizontal axis is the index number, and the ordinate is the eigenvalue, which is the extracted characteristic root.

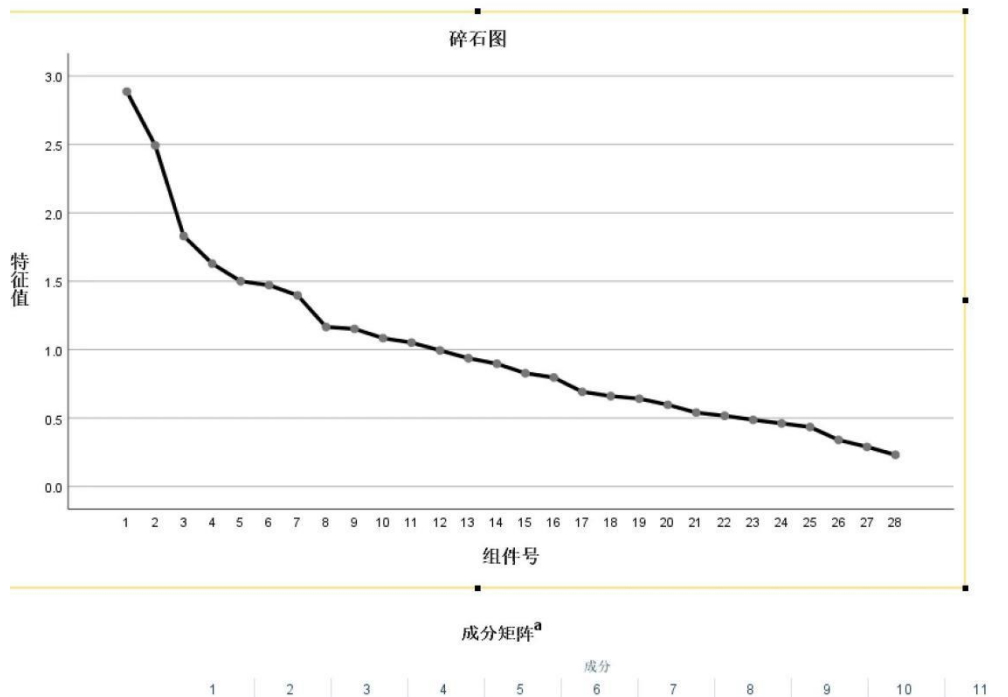


Chart 13

3.3.2 Analysis of customer consumption willingness based on multiple and orderly Logistic regression

In order to further analyze customers' consumption willingness, we used multiple logistics regression to analyze the impact of each factor on customers' willingness to consume milk tea, including hot pot taste, eating habits and other factors.

3.3.3 Variable simulation

According to the questionnaire data, for different questions and data results, we set different variables, and determined all possible values of these variables. According to the data, the marginal percentage is the marginal probability was calculated. For example, with the diet frequency of milk tea in hot pot, we listed the four possible values of never, sometimes, very little and often, and calculated their respective probability statistically. The probability of respondents often having milk tea in hot pot was 25.9%.

个案处理摘要

		个案数	边际百分比
12、如果火锅店旁边开了奶茶店是否会增加喝奶茶的欲望？	是	153	76.1%
	否	48	23.9%
4、(川味火锅)	0	45	22.4%
	有	156	77.6%
4、(广式海鲜火锅)	无	156	77.6%
	有	45	22.4%
4、(京味铜炉火锅)	无	148	73.6%
	有	53	26.4%
4、(苏杭菊花火锅)	无	191	95.0%
	有	10	5.0%
4、(云南滇味火锅)	无	183	91.0%
	有	18	9.0%
4、(浙江八生火锅)	无	190	94.5%
	有	11	5.5%
4、(东北白肉火锅)	无	188	93.5%
	有	13	6.5%
4、(香港“牛肉火锅”)	无	158	78.6%
	有	43	21.4%
4、(上海“什锦火锅”)	无	174	86.6%
	有	27	13.4%
4、(日式寿喜烧火锅)	无	126	62.7%
	有	75	37.3%
4、(韩式部队火锅)	无	130	64.7%
	有	71	35.3%
7、是否有一边吃火锅一边喝奶茶的习惯	从不	42	20.9%
	很少	41	20.4%
	有时	66	32.8%
	经常	52	25.9%
有效		201	100.0%
缺失		0	
总计		201	
子群体		122 ^a	

a. 因变量在 109 (89.3%) 子群体中只有一个实测值。

Table 6

3.3.4 Model results

According to the question of whether the desire of milk tea will increase if the milk tea shops open next to the hot pot restaurant, combining with the data of the choose hot pot taste, we made the chart of parameter estimation value, in which (hot pot) =0 represents not choosing hot pot, (hot pot) =1 represents choosing this kind of hot pot.

First of all, when we compare the value of significance, only the significance data of less than 0.05 is reasonable. Therefore, in the selection of hot pot taste, the data of Shanghai Mixed hot pot and Japanese Shouxi burning hot pot are more reasonable and available. Then we analyzed according to the value of Exp (B), taking spicy hot pot from Sichuan as an example, the willingness of the people who do not choose this hot pot to choose to consume milk tea is 0.415 times the consumption willingness of the people who chose this hot pot. That is, when the respondents chose to eat spicy hot pot from Sichuan, the willingness to consume milk tea is greater. On the other hand, the willingness of people who do not choose seafood hot pot from

Guangdong consume milk tea is 1.614 times that of the willingness of the people who chose this hot pot. That is, when the respondents chose to eat seafood hot pot, the willingness to consume milk tea is smaller. This shows that hot pot taste has a great impact on the consumption willingness of milk tea. Spicy hot pot will increase the willingness to consume milk tea, while lite hot pot will reduce the willingness to consume milk tea.

参数估算值									
12、如果火锅店旁边开了奶茶店是否会增加喝奶茶的欲望？ ^a									
	B	标准 错误	瓦尔德	自由度	显著性	Exp(B)	Exp(B) 的 95% 置信区间		
是	截距	22.377	2.152	108.119	1	.000			
	[4、(川味火锅)=0]	-.880	.533	2.726	1	.099	.415	.146	1.179
	[4、(川味火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(广式海鲜火锅)=0]	.479	.531	.812	1	.367	1.614	.570	4.574
	[4、(广式海鲜火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(京味铜炉火锅)=0]	-.507	.511	.985	1	.321	.602	.221	1.640
	[4、(京味铜炉火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(苏杭菊花火锅)=0]	-17.333	.000	.	1	.	2.967E-8	2.967E-8	2.967E-8
	[4、(苏杭菊花火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(云南滇味火锅)=0]	-.579	.781	.550	1	.458	.561	.121	2.589
	[4、(云南滇味火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(浙江八生火锅)=0]	.359	1.142	.099	1	.753	1.432	.153	13.439
	[4、(浙江八生火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(东北白肉火锅)=0]	-1.590	1.206	1.738	1	.187	.204	.019	2.168
	[4、(东北白肉火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(香港“牛肉火锅”)=0]	.668	.537	1.549	1	.213	1.951	.681	5.587
	[4、(香港“牛肉火锅”)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(上海“什锦火锅”)=0]	-1.872	.833	5.048	1	.025	.154	.030	.787
	[4、(上海“什锦火锅”)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(日式寿喜烧火锅)=0]	1.135	.503	5.094	1	.024	3.113	1.161	8.344
	[4、(日式寿喜烧火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[4、(韩式部队火锅)=0]	-.206	.467	.195	1	.659	.814	.326	2.032
	[4、(韩式部队火锅)=1]	0 ^b	.	0	.	.	.	.	.
	[7、是否有一边吃火锅一边喝奶茶的习惯=1]	-3.266	.672	23.647	1	.000	.038	.010	.142
	[7、是否有一边吃火锅一边喝奶茶的习惯=2]	-1.647	.682	5.831	1	.016	.193	.051	.733
	[7、是否有一边吃火锅一边喝奶茶的习惯=3]	-.232	.670	.120	1	.729	.793	.213	2.946
	[7、是否有一边吃火锅一边喝奶茶的习惯=4]	0 ^b	.	0	.	.	.	.	.

a. 参考类别为：^1。

b. 此参数冗余，因此设置为零。

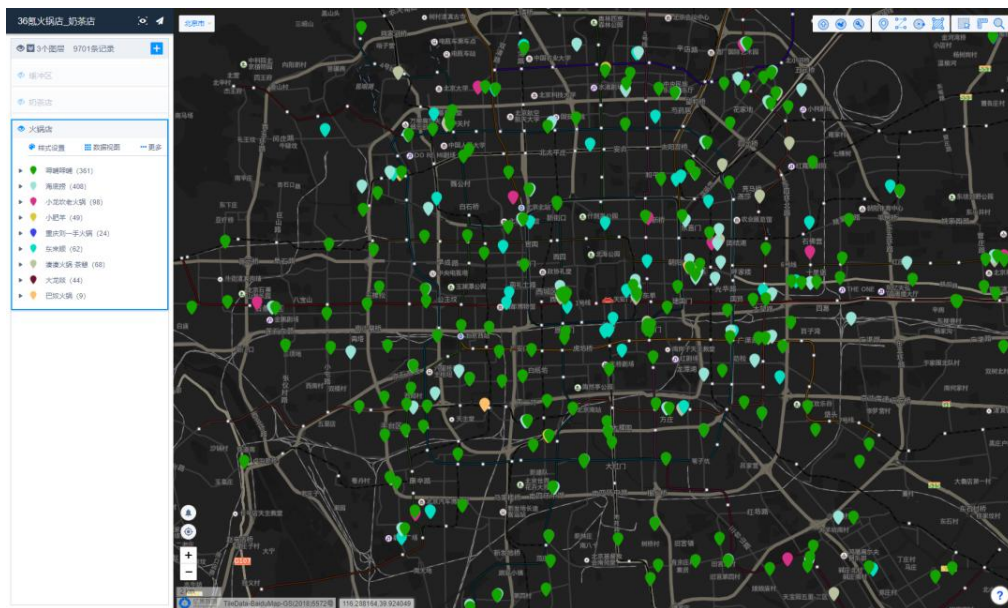
Table 7

4. Models for research

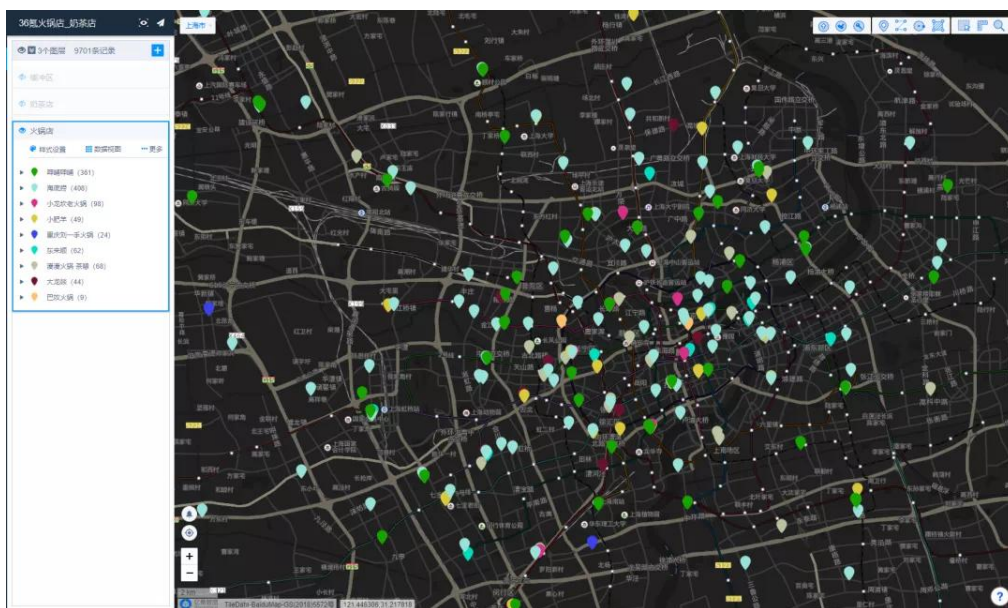
4.1 API map crawler

According to our survey previously, most of interviewees are from Beijing, Shanghai, and southwest part of China. Therefore, we choose 3 typical cities--Beijing, Shanghai, and Chengdu, to see the correlation between hotpot shops and milk tea shops. The 3 standard of cities we choose are as follows, in order to let our analysis become more persuasive. Firstly, the capacity and level of consumption is high. Secondly, cities from northern and southern part, eastern and western part of China should all be included, in order to see the universality. Thirdly, the cities enjoys culture with eating hotpots, or milk tea gains popularity, that means, the consumers in these cities prefer to choose hotpots and milk tea.

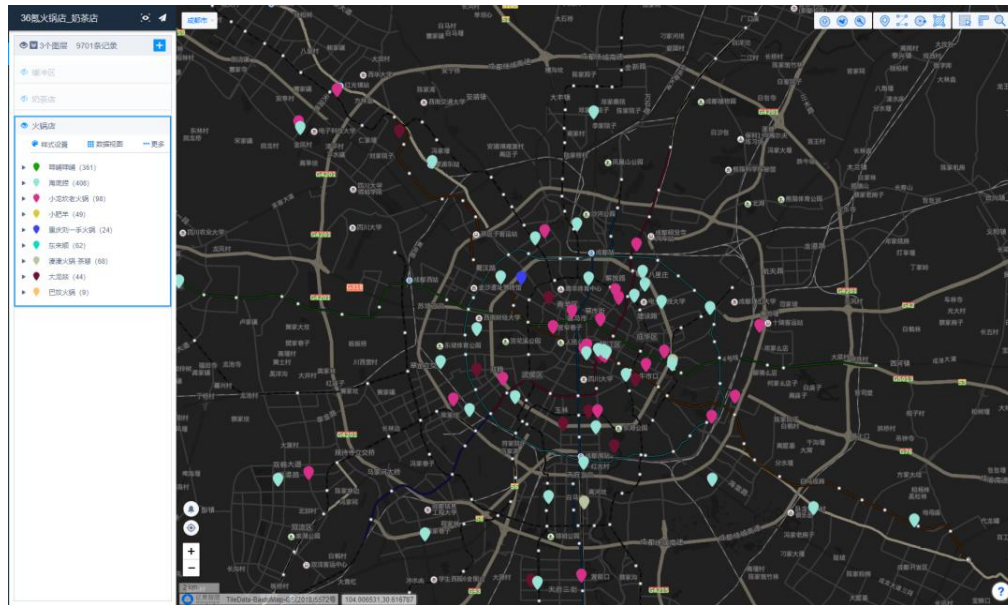
The distribution of hotpot shops, Beijing



Picture 1 source: 36g
The distribution of hotpot shops, Shanghai



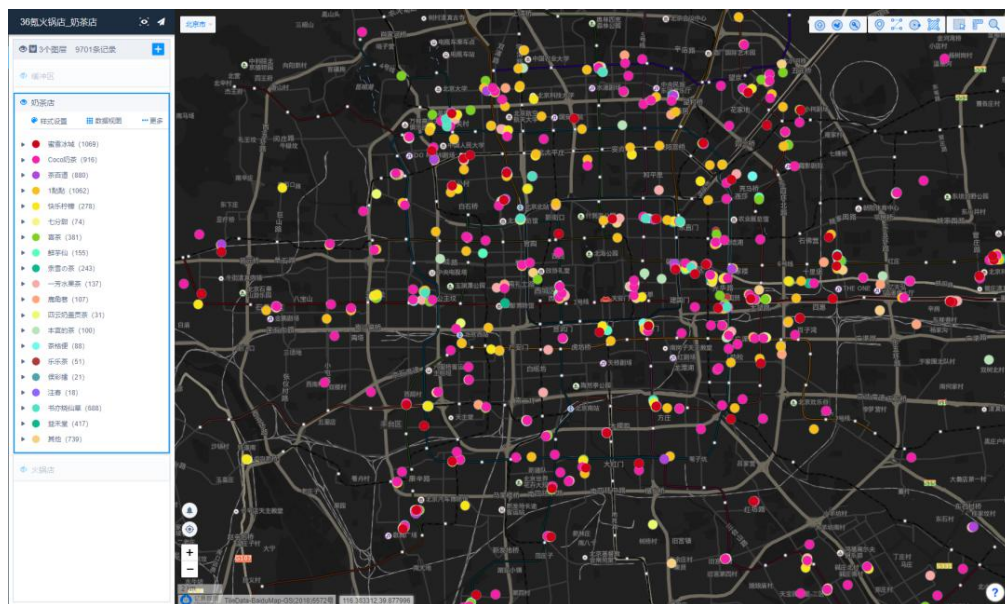
Picture 2 source: 36g
The distribution of hotpot shops, Chengdu



Picture 3 source: 36g

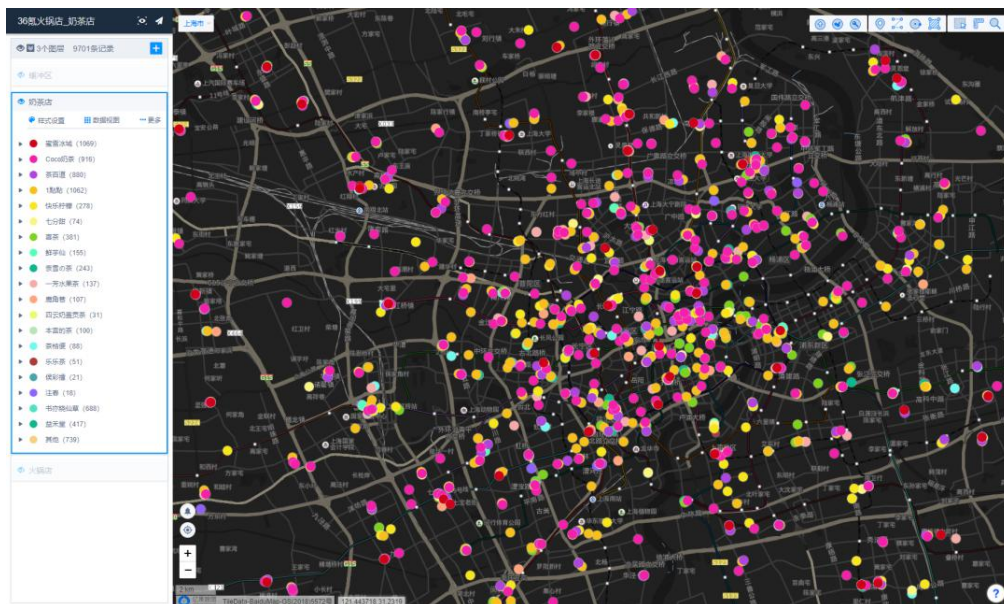
Then, we add the positions of these popular chain milk tea shops around hotpot shops within 500m, including Mixue Ice Cream & Tea, Coco, Cha Baidao, and Yi Diandian.

The distribution of milk tea shops, Beijing

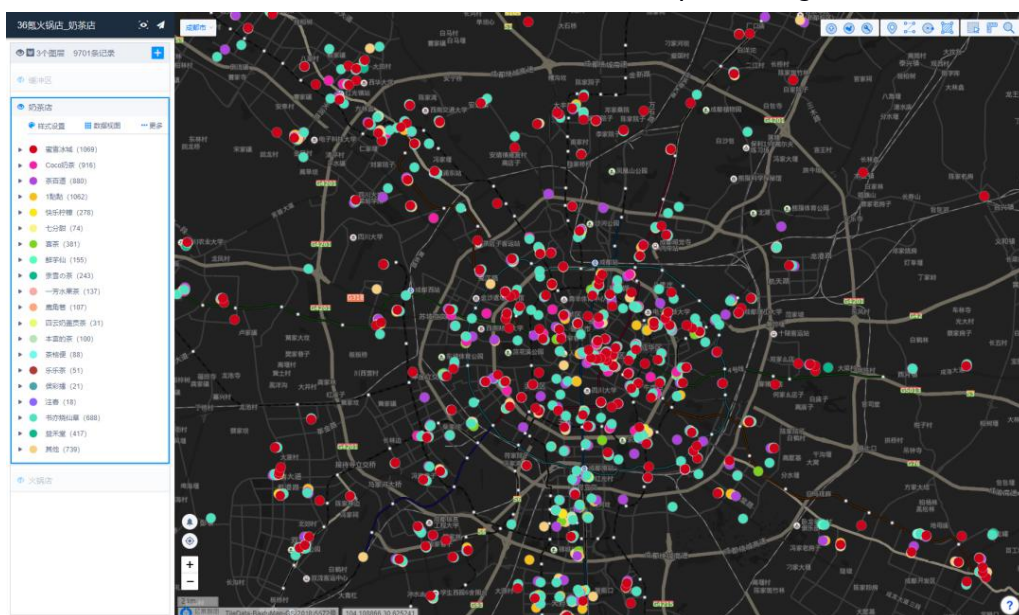


Picture 4 source: 36g

The distribution of milk tea shops, Shanghai



Picture 5 source: 36g
The distribution of milk tea shops, Chengdu



Picture 6 source: 36g

4.2 frequency distribution

By calculating the frequency of milk tea shops that are around hot pot shops within 500m, and the number of milk tea shops in each city, the percentage of milk tea shop near hot pot shops within 500m are as follows.

City	Name of milk tea shops	Milk tea shop around hot pot shops	Total milk tea shops in this city	percentage
Beijing	Coco	133	358	42%
	Mixue Ice Cream & Tea	53	245	22%
	Yi Diandian	57	113	43%

	Cha Baidao	19	135	14%
Shanghai	Coco	168	403	42%
	Mixue Ice Cream & Tea	30	255	12%
	Yi Diandian	60	334	18%
	Cha Baidao	70	243	29%
Chengdu	Coco	14	171	8%
	Mixue Ice Cream & Tea	91	599	15%
	Yi Diandian	24	111	21%
	Cha Baidao	43	276	16%

Table 8

In Beijing, nearly half of Coco and Yi Diandian shops are around hot pot shops. In Shanghai, over 40% of Coco shops are around hot pot shops, followed by Cha Baidao, with the percentage of nearly 30%. In Chengdu, Yi Diandian shops are more likely to set near hot pot shops.

4.3 Linear Regression model

By using the method of single linear regression model, we listed all the combinations of each hot pot shops and each milk tea shops, whose significance is beyond 0.6, since the correlation between 0.6 to 0.8 means strong correlation.

City	Milk Tea	Hot Pot	correlation
Beijing	Coco	Xiabu Xiabu	0.71
	Yi Dian Dian	Xiabu Xiabu	0.73
	Mixue Ice Cream & Tea	Xiabu Xiabu	0.61
	Coco	Haidilao hot pot	0.61
Shanghai	Coco	Haidilao hot pot	0.76
	Cha Bai Dao	Haidilao hot pot	0.61
	Coco	Xiabu Xiabu	0.63
	Coco	Cou Cou	0.62
Chengdu	Mixue Ice Cream & Tea	Haidilao hot pot	0.6
	Yi Diandian	Xiao Long Kan	0.64

Table 9

We see that in Beijing, the correlation between Yi Diandian and Xiabu Xiabu can reach 0.73. Coco shops have strong correlations with both Xiabu Xiabu. In Shanghai, Coco shops have strong relationship with Haidilao, Cou Cou,

and Xiabu Xiabu. Cha Baidao shops also have correlation with Haidilao. In Chengdu, Mixue Ice Cream & Tea and Haidilao, Yi Diandian and Xiao Longkan have strong correlation.

Therefore, there's strong relationship between typical chain hot pot shops and milk tea shops. Therefore, we may infer that setting the milk tea shops near hot pot shops may be a strategy for some most popular milk tea brands. In Beijing, alliances business of Yi Diandian are more likely to gain profit, if they set their locations near Xiabu Xiabu, so do Coco. In Shanghai, franchises of Coco may be a wise choice to run their business near Haidilao.

4.4 Marginal cost & marginal model for saturation

However, if the competitors are too much near hot pot shops, the effect brought from customers may not be significant. To know at most how much milk teas shops should be set up near hot pot shops, under the circumstance of fully used customer volume. According to Nash equilibrium, the optimal outcome of a game is one where no player has an incentive to deviate from their chosen strategy after considering an opponent's choice. We assume that new entrants will share the profit equally. Assuming that the customer volumes from hotpot shops are unchanged, our team is going to find an optimal point by using the model between marginal benefit and marginal cost. According to our previous survey, interviewees who always, or sometimes, or seldom eat hotpots with milk tea is 79.1%. According the research online, the average customer volumes in hotpot shops in upper cities is about 300. Since the frequency of drinking milk tea with hotpots has influence on the customer volumes, our team use weighted average to calculate the amount of people who will order milk tea. We value people who 'always' drink while eating hotpots as 3, who 'sometimes' drink while eating hotpots as 2, who 'seldom' drink as 1, then the weighted average of people who drink milk tea is 98.4. Therefore, we assume that the customer volumes from hotpot shops to milk tea shops is 125 per day. Doing the survey of the prices and costs of milk tea, we assume that valuing marginal benefit per cup as 15 yuan, and marginal cost as 30 yuan is reasonable. Hence, we can see the optimal cost.

Argument meaning	assignment
Whether drink milk tea while eating hotpot	Always =3, sometimes=2, seldom=1, never=0

Table 10

For 6.7 shops, the marginal cost is equal to marginal benefit, which is the optimal point. Therefore, at most 6 milk tea shops is limited around hotpot shops within 500m.

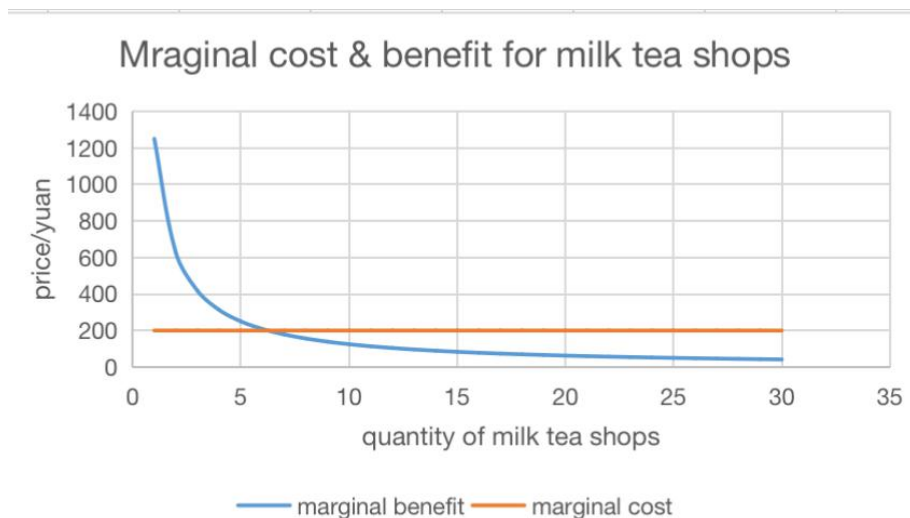


Chart 13

5. Conclusion

According to our survey, there are huge demand of consumers to eat hot pot with milk tea, which means that there's a great potential for milk tea brand, especially franchises to set up their base near hot pot shops. The locations give references for milk tea shop to find new ways to gain profit. The outcome of questionnaires shows that compared with elder people or mid-aged people who focus more on health and never buy milk tea while having hotpot, young are more likely to pay for the bill of milk tea. Therefore, customers from 18-30 are the target groups. As for the taste, people who eat spicy hot pot are more likely to buy milk tea while people eat hot pot with bland taste are not. What new entrants can do is to make promotion to attract young consumers. Or to cooperate with hot pot shops. Another idea is to create new products that are healthy and bland, and cater to middle-aged people or elderly.

Secondly, we find that the distance between milk tea shops and hot pot shops have strong correlations, especially in some magcities, such as Beijing. Therefore, when the cost of transport and materials as well as rent do not have significant influence, setting new milk tea shop within 500m around hot pot shops may be a wise choice. For different brands, it is important to find a hot pot shops with strong correlations, like Yi Diandian shops set their locations near Xiabu Xiabu in Beijing. Coco shops set their locations near Haidilao in Shanghai.

Thirdly, to avoid the fierce competition for each milk tea brand, and reduce the customer volumes for each milk tea shop, then drop the revenue, we suggest that 6 milk tea shops are maximum within 500m. If a shop make some drinks in vogue that way better than other shops, perhaps it has absolute competence. However, it is unrealistic in the market.

Since the hotpot industry and milk tea industry witness a bloom, and the location of new shops to attract customers may be essential, our research

may provide companies a practical strategy.

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Appendix

1.您的性别

男

女

2.请选择城市:

3.年龄

10-18

18-30

30-50

51-70

70 以上

4.吃火锅的偏好

川味火锅

广式海鲜火锅

京味铜炉火锅

苏杭菊花火锅

云南滇味火锅

浙江八生火锅

东北白肉火锅

香港“牛肉”火锅

上海“什锦”火锅

日本寿喜烧火锅

韩式部队火锅

5.吃火锅的频率

经常

有时

很少

从不

6.吃火锅的具体频率

一周一次

一周三次

一周三次以上

一月一次

一月三次

一年一次

一年三到五次

一年五次以上

7.是否有一边吃火锅一边喝奶茶的习惯

经常（跳转到第八题）

有时（跳转到第八题）

很少（跳转到第八题）

从不

8.在吃火锅的时候喝奶茶的原因

解渴解辣

味道好喝

性价比高

9.一般喝奶茶的价位

10 元以内

10-20

21-30

30 以上

10.偏好外卖还是到店自取

到店自取

外卖

11.火锅店和奶茶店距离多远会选择自取

隔壁店铺

楼下或楼上

步行五分钟以内

1km 以内

2km 以内

5km 以内

10km 以内

12.如果火锅店旁边开了奶茶店是否会增加喝奶茶的欲望

是

否